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WEEK NO.	PROBLEMS WITH DESCRIPTION		PAGE NO.	SIGNATURE OF THE TEACHER WITH DATE
1	1#	Discuss the process by which Python executes a program, focusing on the stages involved from source.		
	2#	Discuss the fundamental features of the Python programming language that make it widely used for modern software development.		
	3#	Install IDLE interpreter for Python programming by following the mentioned steps.		
2	1#	Use interactive Shell to print: User said "Great app!" and directory paths.		
	2#	A shopkeeper uses Python shell for quick billing calculations. Evaluate various arithmetic expressions (addition, subtraction, multiplication, division, modulus).		
	3#	A college form stores student names. Create variables firstName and lastName with your own names.		
	4#	Print firstName and lastName in a single line using f-strings.		
	5#	A heritage museum kiosk needs an automated display-and-billing script. Create a single Python file kioskSystem.py that performs billing calculations and displays information messages.		
	6#	An engineer calculates load using a polynomial. Given specific variable values, compute the result of the algebraic expression.		
	7#	Write a program to compute the value of a specific algebraic expression where x and y are read using the input() function.		
	8#	A technician optimizes multiplication using bitwise operations. Read an integer and multiply it by 2 using the left shift operator (<<).		
3	1#	A bookstore's billing app needs a quick utility script. Write a Python program that reads N integers (prices) and computes: Sum, Product, and Average.		
	2#	A weather station needs temperature conversion. Write a program to convert Celsius to Fahrenheit.		

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	3#	An architecture tool calculates dimensions. Write a program to compute the Surface Area and Volume of a Cuboid given length, breadth, and height.		
	4#	Write a program to add two numbers without using arithmetic operators, using bitwise operators.		
	5#	A hardware technician tests divider circuits. Read an integer and divide it by 4 using the bitwise >> operator.		
	6#	A robotics controller swaps sensor IDs efficiently. Read two integers and exchange their values using the XOR (^) operator.		
	7#	A gaming console optimizes score calculation. Multiply a number by 10 using bitwise operators.		
	8#	A data-logging device swaps configuration settings. Read two integers and exchange their values using addition and subtraction (avoid using a third variable).		
	9#	A biology simulation needs repeated multiplication. Write a program to compute the factorial of an integer using a loop.		
	10#	A math module checks sequence validity. Write a program to determine whether a number is a Fibonacci number.		
	11#	A security system verifies prime-number-based keys. Write a program to check whether an integer is prime, or else output its first factor.		
	12#	A signal-processing unit finds harmony between two frequencies. Write a program to compute the GCD of two numbers.		
4	1#	A teacher wants to automate small mathematical tasks. Define functions to: compute average of marks, convert temperature, and calculate perimeter of a rectangle.		
	2#	A science lab technician is calculating the volume of spherical containers. Write a function that accepts the radius and returns the volume of the sphere.		
	3#	A landscaping company maintains circular flower beds. Write a function that takes outer radius R and inner radius r, validates them, and returns the effective usable area.		

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	4#	A data analyst needs factorial values to compute permutations. Write a Python function to calculate the factorial of a non-negative integer.		
	5#	An accountant wants a utility that extracts a digit-sum check value. Write a Python function that returns the sum of digits of a given integer.		
	6#	A math research assistant is verifying test values. Write a Python function to check whether a number is a Perfect Number.		
	7#	Write a Python function to check whether a given number is a prime number or not.		
	8#	A text-analysis tool needs statistics from a sentence. Write a function that accepts a string and counts the number of uppercase and lowercase letters.		
5	1#	A data automation team is creating a utility module. Write a program that accepts inputs and uses functions to: find largest of three numbers, compute volume of shapes, area of rectangle, circumference of circle, swap variables, and find distance between points.		
	2#	A billing system receives an unknown count of numeric entries. Write a function that accepts arbitrary integers (*args) and returns their sum.		
	3#	A text-filtering tool needs to remove noisy characters. Write a program that takes a string and removes all characters at even index values.		
	4#	Write a Python script that takes input from the user and displays that input back in upper and lower cases.		
	5#	A simple analytics module must calculate keyword frequency. Write a program to count occurrences of each word in a sentence.		
	6#	A document cleanup tool needs to normalize text. Write a program to remove indentation from each line of a multi-line text.		
	7#	Write a function that reverses a string by modifying the input array in-place with O(1) extra memory.		
	8#	Given a string, reverse the order of characters in each word within a sentence while preserving whitespace and word order.		

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	9#	A parser converts messages into tokens. Write a program to convert a string into a list of characters.		
	10#	Write a Python program to count and display all vowels present in the text, as well as consonants and their count.		
6	1#	A grading system compares three internal-assessment scores. Write a program to find the highest score among the three using only if statements.		
	2#	A sensor module collects readings from three probes. Write a program to determine the lowest reading using if–else logic.		
	3#	Write a program to check if a year is a leap year.		
	4#	Write a program to print the number of days in a month using chained conditionals.		
	5#	Write a program that takes a month and year, and prints the correct number of days, accounting for leap years.		
	6#	An architectural software checks triangular panels. Write a program to compute area and identify if a triangle is equilateral, isosceles, or scalene.		
	7#	A device logs data indexes. Given two numbers r1 and r2, create a list of integers from r1 to r2 using range().		
	8#	A parser receives a mixed list. Write a program to extract and add only numeric items using isinstance().		
	9#	A game engine verifies level IDs. Write a program to check whether a given list of numbers is consecutive.		
	10#	FizzBuzz: Given integer n, return a string array where multiples of 3 are "Fizz", 5 are "Buzz", and both are "FizzBuzz"		
7	1#	A cryptographic security module uses "Narcissistic Numbers". Write a program using a while loop to check if a numeric key is an Armstrong number.		
	2#	A biological simulation tool models cell growth. Write a recursive function to generate the first n terms of the Fibonacci sequence.		

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	3#	A data science team needs to calculate permutations. Write a recursive function calculate_routes(n) to compute the factorial of a number.		
	4#	A telecommunications engineer synchronizes signals. Write a recursive function find_signal_sync(a, b) to find the GCD using the Euclidean algorithm.		
	5#	Write a program to print a specific inverted right-angle triangle pattern using a for loop.		
	6#	A network packet analyzer receives packet IDs. Filter these into two lists: even IDs (low priority) and odd IDs (high priority).		
	7#	Given an integer x, return true if x is a palindrome, and false otherwise.		
	8#	An inventory management system consolidates records. Merge two lists of product IDs and display the final list in sorted order.		
	9#	An algorithm visualization tool demonstrates sorting. Write a program to find the second largest number in a list using the Bubble Sort algorithm.		
8	1#	A file management system needs to organize filenames. Write a program to sort a list of strings based on string length.		
	2#	A marketing team has two email lists. Create a master list containing all unique emails from both lists (Union).		
	3#	A cyber-security tool compares server logs. Find the common IP addresses between two lists (Intersection).		
	4#	A data visualization tool needs to plot a curve. Generate a list of tuples where the first element is a number and the second is five times its cube.		
	5#	An NLP pre-processor needs to clean text. Use string methods to tokenize, normalize, find index, search for characters, and sanitize text.		
	6#	A data compression utility analyzes entropy. Calculate the frequency of every character present in a given string.		
	7#	A simple encryption tool obfuscates passwords. Write a program to reverse the order of characters in		

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		a string.		
	8#	A product labeling system generates "SKU codes". Generate a new string made of the first 2 and last 2 characters of a product name.		
	9#	A library catalog system contains sorted Book IDs. Implement Binary Search to find a specific Book ID.		
	10#	An educational algorithm visualizer needs to show sorting steps. Implement Insertion Sort and print the state of the list after every iteration		
9 & 10	1#	A financial risk analysis tool needs to determine volatility. Identify the Maximum and Minimum values in a list of daily closing prices.		
	2#	An inventory checking system allows a store manager to scan a product ID. Write a program to search for a specific element in a list of product IDs.		
	3#	A music playlist app needs to organize songs. Sort a raw list of song titles alphabetically.		
	4#	An election tally system has a dictionary of votes. Sort the dictionary to display the winner and the candidate with the fewest votes.		
	5#	A Data Science team is compiling a "Complete Patient Record". Write a script to concatenate three distinct dictionaries into one single Patient Profile.		
	6#	An embedded system needs to sort a small buffer. Implement the Optimized Bubble Sort algorithm (using a flag mechanism).		
	7#	A Big Data processing engine needs to sort a massive log file. Implement the Merge Sort algorithm using recursion.		
	8#	A real-time gaming engine needs a high-speed sorting algorithm. Implement the Quick Sort algorithm.		
11 & 12	1#	A DevOps engineer manages server configuration. Write a script to inspect, modify, clean up, and sort a dictionary of settings.		
	2#	A linguistic research tool analyzes character distribution. Create a Histogram (dictionary) that counts how often each letter appears in a string.		
	3#	A corporate security system stores employee data.		

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		Write a program to perform a Reverse Lookup (find Key by Value) for a badge ID.		
	4#	An e-commerce pricing engine applies a "Flash Sale". Merge a Sale_Updates dictionary into a Catalog_Prices dictionary, overwriting existing values.		
	5#	A Math Learning App needs a lookup table. Generate a dictionary where keys are integers 1-15 and values are their squares.		
	6#	Implement the Matrix Chain Multiplication problem using dynamic programming.		
	7#	A GPS Navigation System calculates routes. Find the Shortest Path in a weighted graph using Dijkstra's Algorithm.		
13 & 14	1#	A government agency connects remote islands. Implement Kruskal's Algorithm to find the Minimum Spanning Tree (MST).		
	2#	A logistics company uses a delivery drone. Implement the Traveling Salesman Problem (TSP) to find the shortest route visiting all locations.		
	3#	A museum security team places guards at intersections. Approximate a solution to the Vertex Cover Problem using a Greedy Algorithm.		